

DATA MINING LAB

Midterm Lab Exam

Disease: Malaria

Submitted by Group 9 :

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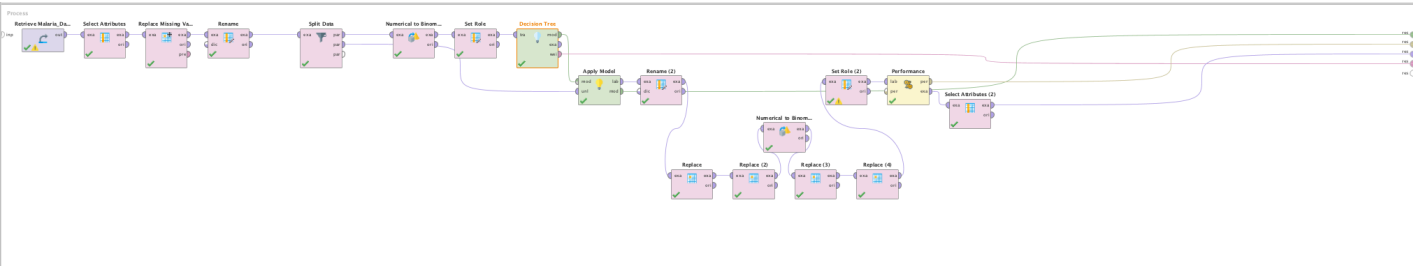
October 30, 2025

DATA DICTIONARY:

Column Name	Description	Data Type	Example Value	Remarks
Age	Represents the patient's age in years.	Integer	52	Helps identify age-related malaria risk.
Fever	Indicates if the patient experienced fever (1 = Yes, 0 = No).	Integer (0/1)	0	Fever is a common symptom of malaria.
Headache	Indicates if the patient reported a headache (1 = Yes, 0 = No).	Integer (0/1)	0	Often occurs with malaria infection.
Abdominal_Pain	Indicates if the patient had abdominal pain (1 = Yes, 0 = No).	Integer (0/1)	0	Can be associated with severe malaria.
General_Body_Malaise	Indicates if the patient felt general body weakness or fatigue (1 = Yes, 0 = No).	Integer (0/1)	1	A common symptom during malaria infection.
Dizziness	Indicates if the patient experienced dizziness (1 = Yes, 0 = No).	Integer (0/1)	0	May occur due to fever or dehydration.
Vomiting	Indicates if the patient experienced vomiting (1 = Yes, 0 = No).	Integer (0/1)	0	Common in moderate to severe malaria cases.
Confusion	Indicates if the patient showed confusion or disorientation (1 = Yes, 0 = No).	Integer (0/1)	0	Seen in severe malaria or cerebral cases.
Backache	Indicates if the patient complained of back pain (1 = Yes, 0 = No).	Integer (0/1)	1	A possible symptom of infection or weakness.
Chest_Pain	Indicates if the patient reported chest pain (1 = Yes, 0 = No).	Integer (0/1)	0	May occur due to muscle fatigue or infection.

Coughing	Indicates if the patient experienced coughing (1 = Yes, 0 = No).	Integer (0/1)	0	Not a core malaria symptom but may appear in co-infections.
Joint_Pain	Indicates if the patient had joint pain (1 = Yes, 0 = No).	Integer (0/1)	0	Can be a sign of body inflammation due to infection.
Target	Indicates malaria diagnosis (1 = Positive, 0 = Negative).	Integer (0/1)	0	Used as the prediction label in model training.

DESIGN:



PROCESS DIAGRAM: The process retrieves, cleans, and prepares the malaria dataset, applies a Decision Tree model to predict malaria risk, then renames and replaces output values for better readability before evaluating the model's performance.

SELECT ATTRIBUTES:

Parameters

Select Attributes

type

include attributes

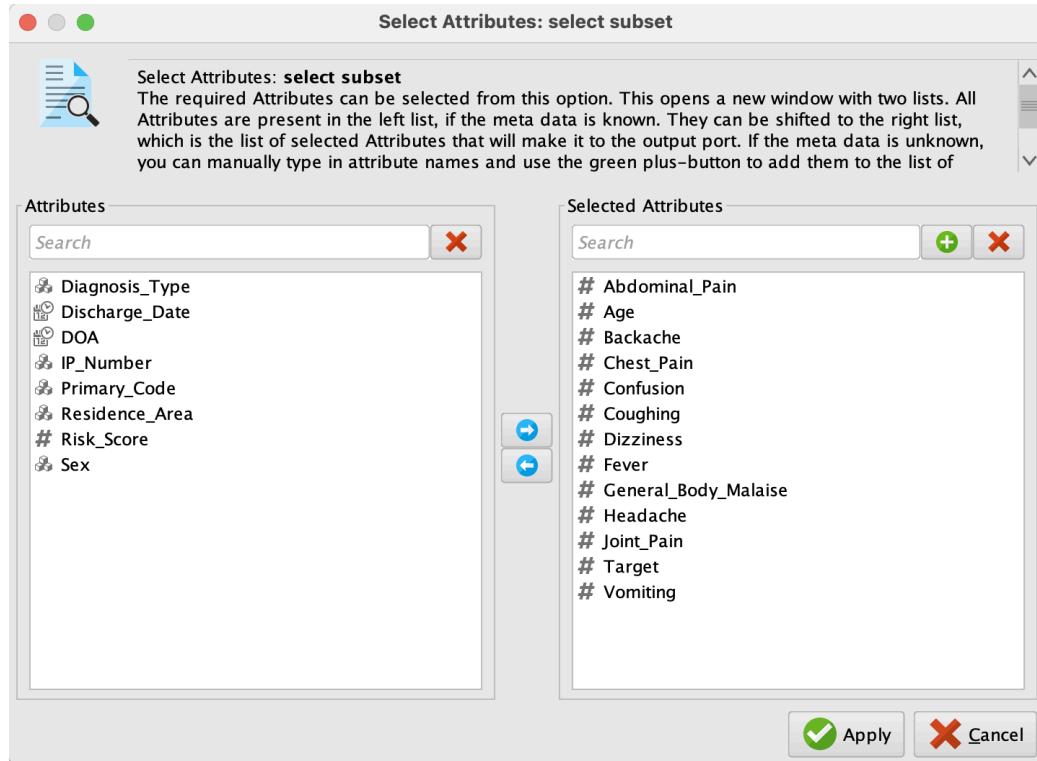
attribute filter type

a subset

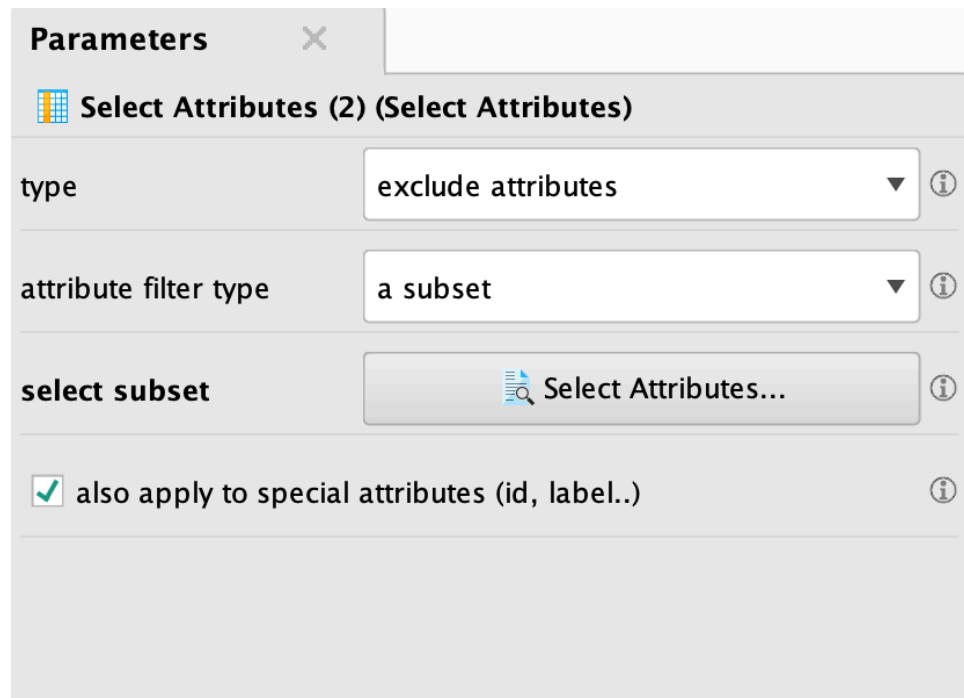
select subset

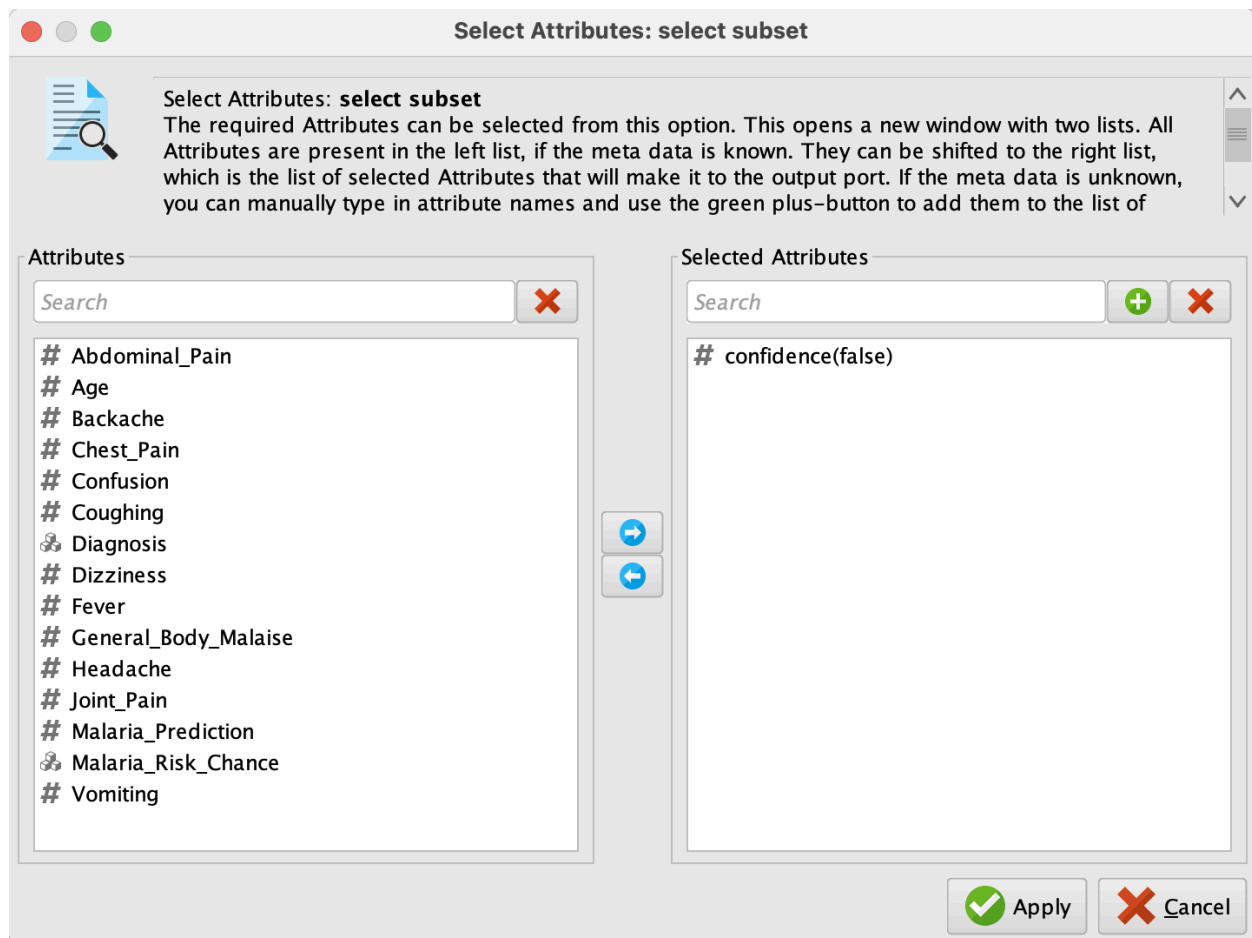
Select Attributes...

☐ also apply to special attributes (id, label..)



SELECT ATTRIBUTES: Used to select the important attributes that will be used for training and testing the model.





SELECT ATTRIBUTES: Used to exclude the confidence(false) column because it is not needed for the final results table of our Malaria data.

RENAME ATTRIBUTES:

Edit Parameter List: rename attributes



Edit Parameter List: **rename attributes**

This parameter is available if and only if the *dictionary input* port is not connected. To rename one or more attributes click on the "Edit List (0)..." button. Here you can select attributes and assign new names to them.

old name	new name
Target	Diagnosis

 Add Entry

 Remove Entry

 Apply

 Cancel

Rename 1: Renaming column title "Target" to "Diagnosis" for better data representation

Edit Parameter List: rename attributes



Edit Parameter List: **rename attributes**

This parameter is available if and only if the *dictionary input* port is not connected. To rename one or more attributes click on the "Edit List (0)..." button. Here you can select attributes and assign new names to them.

old name		new name
confidence(true)	▼	Malaria_Risk_Chance
prediction(Diagnosis)	▼	Malaria_Prediction



Add Entry



Remove Entry



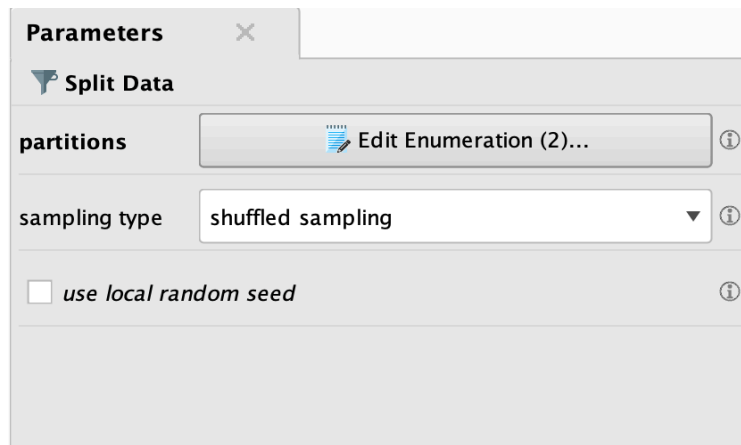
Apply



Cancel

Rename 2: Renaming column title "confidence(true)" (confidence of the model that the patient has malaria) to "Malaria_Risk_Chance" for better data representation and "prediction(Diagnosis)" (model's prediction if the patient has Malaria) to "Malaria_Prediction"

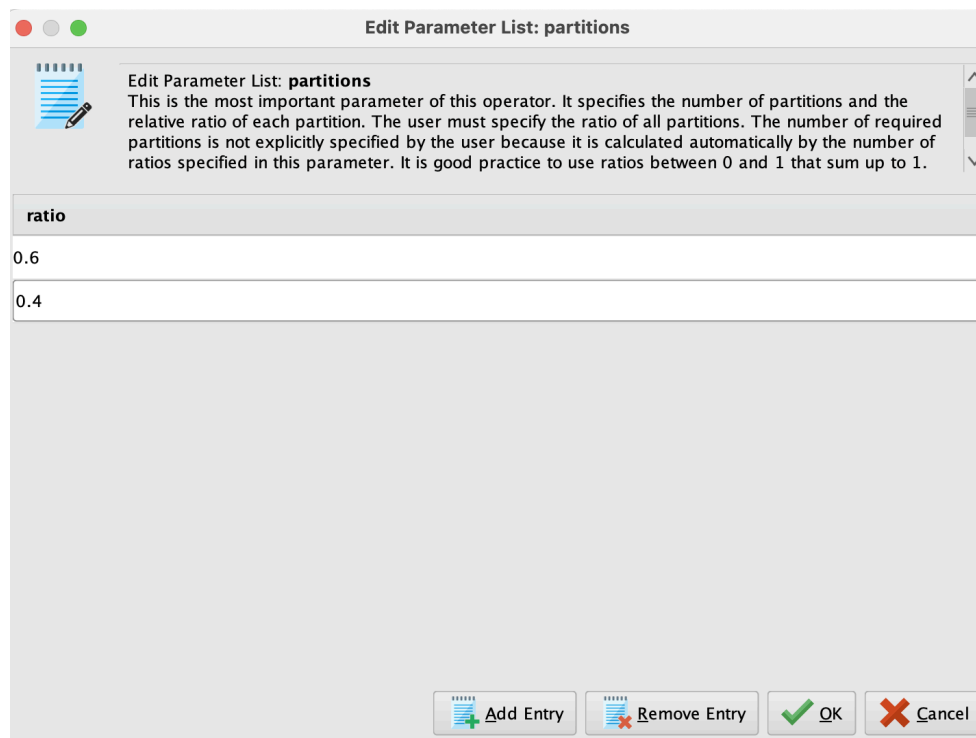
SPLIT DATA:



The 'Parameters' dialog for the 'Split Data' operator is shown. It has a title bar with a close button. Below the title bar is a section header 'Split Data' with a funnel icon. The parameters are:

- partitions**: A button labeled 'Edit Enumeration (2)...' with an information icon.
- sampling type**: A dropdown menu set to 'shuffled sampling' with an information icon.
- use local random seed**: A checkbox that is currently unchecked, with an information icon.

Split Data: Makes the training and testing sets have a fair mix of different kinds of cases by randomly mixing (shuffling) all the rows.



The 'Edit Parameter List: partitions' dialog is shown. It has a title bar with standard window controls. The main area contains a text box with the following text:

Edit Parameter List: partitions
This is the most important parameter of this operator. It specifies the number of partitions and the relative ratio of each partition. The user must specify the ratio of all partitions. The number of required partitions is not explicitly specified by the user because it is calculated automatically by the number of ratios specified in this parameter. It is good practice to use ratios between 0 and 1 that sum up to 1.

Below the text box is a table with the following data:

ratio
0.6
0.4

At the bottom of the dialog are four buttons: 'Add Entry' (with a plus icon), 'Remove Entry' (with a minus icon), 'OK' (with a checkmark icon), and 'Cancel' (with an X icon).

Partitions: 60-40 split was used to ensure the model learns from sufficient data while having enough reliable data for testing.

REPLACE MISSING VALUES:

Parameters

 Replace Missing Values

attribute filter type all

☐ invert selection

☐ include special attributes

default average

columns Edit List (14)...

Edit Parameter List: columns



Edit Parameter List: **columns**

Different Attributes can be provided with a different type of replacements through this parameter. The default function selected by the *default* parameter is applied on Attributes that are not explicitly mentioned in the *columns* parameter.

attribute	replace with
Age	average
Fever	zero
Headache	zero
Abdominal_Pain	zero
General_Body_Malaise	zero
Dizziness	zero
Vomiting	zero
Confusion	zero
Backache	zero
Chest_Pain	zero
Coughing	zero
Joint_Pain	zero
Target	none
Risk_Score	average

 Add Entry

 Remove Entry

 Apply

 Cancel

Edit Parameter List: Replacing the missing values in the dataset to the most appropriate default value

REPLACE:

Parameters

Replace

attribute filter type

single

attribute

Malaria_Prediction

☐ invert selection

☒ include special attributes

replace what

true

replace by

Positive

Replace 1: Replacing all values labeled “true” to “Positive” in the Malaria_Prediction column for better data representation.

Parameters

Replace (2) (Replace)

attribute filter type

single

attribute

Malaria_Prediction

☐ invert selection

☒ include special attributes

replace what

false

replace by

Negative

Replace 2: Replacing all values labeled “false” to “Negative” in Malaria_Prediction column for better data representation

Parameters ×

 **Replace (3) (Replace)**

attribute filter type single ▼ ⓘ

attribute Diagnosis ▼ ⓘ

☐ invert selection ⓘ

☒ include special attributes ⓘ

replace what true  ⓘ

replace by Positive ⓘ

Replace 3: Replacing all values labeled “true” to “Positive” in the Diagnosis column for better data representation.

Parameters ×

 **Replace (4) (Replace)**

attribute filter type single ▼ ⓘ

attribute Diagnosis ▼ ⓘ

☐ invert selection ⓘ

☐ include special attributes ⓘ

replace what false  ⓘ

replace by Negative ⓘ

Replace 4: Replacing all values labeled “false” to “Negative” in Diagnosis column for better data representation

NUMERICAL TO BINOMIAL:

Parameters

 **Numerical to Binominal**

attribute filter type

single

attribute

Diagnosis

☐

invert selection

☐

include special attributes

min

0.0

max

0.0

Parameters

 **Numerical to Binominal (2) (Numerical to Binominal)**

attribute filter type

single

attribute

Diagnosis

☐

invert selection

☐

include special attributes

min

0.0

max

0.0

Parameter: Used to convert numerical values into two categories so the model can treat them as classification labels

SET ROLES:

Edit Parameter List: set roles

 **Edit Parameter List: set roles**
This parameter is used to set the roles of one or more Attributes. A click on "Edit List (0)..." opens a menu with Attribute name and target role pairs. The role of an Attribute will be changed to the given target role for each of the pairs. Following target roles are possible:

attribute name	target role
Diagnosis	label

 Add Entry  Remove Entry  Apply  Cancel

Edit Parameter List: set roles

 **Edit Parameter List: set roles**
This parameter is used to set the roles of one or more Attributes. A click on "Edit List (0)..." opens a menu with Attribute name and target role pairs. The role of an Attribute will be changed to the given target role for each of the pairs. Following target roles are possible:

attribute name	target role
Diagnosis	label

 Add Entry  Remove Entry  Apply  Cancel

Set Roles: Defines the purpose of the Diagnosis attribute as the label to predict

PERFORMANCE:

Parameters

%

Performance (Performance (Classification))

main criterion

first

i

☒

accuracy

i

☐

classification error

i

☐

kappa

i

☐

weighted mean recall

i

☐

weighted mean precision

i

☐

spearman rho

i

☐

kendall tau

i

☐

absolute error

i

☐

relative error

i

☐

relative error lenient

i

☐

relative error strict

i

☐

normalized absolute error

i

☐

root mean squared error

i

☐

root relative squared error

i

☐

squared error

i

☐

correlation

i

☐

squared correlation

i

☐

cross-entropy

i



[Hide advanced parameters](#)

Default: Checks the accuracy of the model's predictions

RESULTS:

Open in

Turbo Prep

Auto Model

Interactive Analysis

Filter (649 / 649 examples): all

Row No.	Diagnosis	Malaria_Prediction	Malaria_Risk_Chance	Age	Fever	Headache	Abdominal...	General_Bo...	Dizziness	Vomiting	Confusion	Ba
1	Negative	Negative	0	52	0	0	0	1	0	0	0	1
2	Positive	Positive	0.995	75	1	0	1	1	1	0	1	0
3	Positive	Positive	0.995	30	1	1	1	1	0	0	1	1
4	Positive	Positive	1	62	0	1	0	1	0	1	1	0
5	Positive	Positive	1	78	0	1	1	1	1	1	1	1
6	Positive	Positive	1	48	0	1	1	0	1	0	1	1
7	Negative	Negative	0	78	0	0	0	1	0	0	1	0
8	Positive	Positive	1	15	1	1	1	0	0	0	1	0
9	Positive	Positive	0.995	28	1	1	1	1	1	1	0	1
10	Negative	Negative	0	3	0	1	1	0	0	1	0	0
11	Positive	Positive	1	58	1	1	1	0	1	0	0	0
12	Positive	Negative	0.143	16	0	0	1	0	1	1	0	0
13	Positive	Positive	1	36	0	1	0	0	0	1	1	0
14	Positive	Positive	0.995	18	1	1	1	1	1	0	1	0
15	Positive	Positive	1	68	0	1	0	1	1	1	1	1
16	Positive	Positive	1	33	1	1	0	0	1	0	1	1
17	Positive	Positive	1	51	1	1	0	0	1	0	1	0
18	Positive	Positive	1	69	1	0	1	0	1	0	1	0
19	Positive	Positive	0.995	8	1	1	1	1	1	0	0	0
20	Positive	Positive	1	47	0	0	1	1	1	0	1	1
21	Positive	Positive	0.995	17	1	0	0	1	1	0	1	0
22	Positive	Positive	1	22	0	0	0	1	1	1	0	0
23	Positive	Positive	1	69	0	1	1	1	1	0	1	0
24	Positive	Positive	1	84	1	1	1	0	1	0	0	0
25	Positive	Positive	0.833	12	0	0	0	1	1	0	1	0
26	Positive	Positive	1	24	1	0	0	0	1	1	1	0

ExampleSet (649 examples,3 special attributes,12 regular attributes)

Results: The table shows the output of our malaria prediction model, listing 649 patient cases with their predicted diagnosis, malaria risk and probability scores, and corresponding symptom data.

DECISION TREE PARAMETERS:

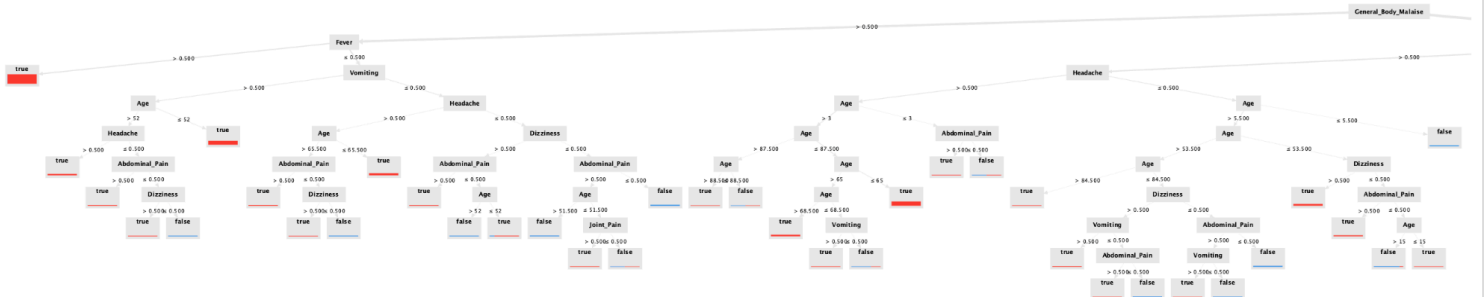
Parameters

Decision Tree

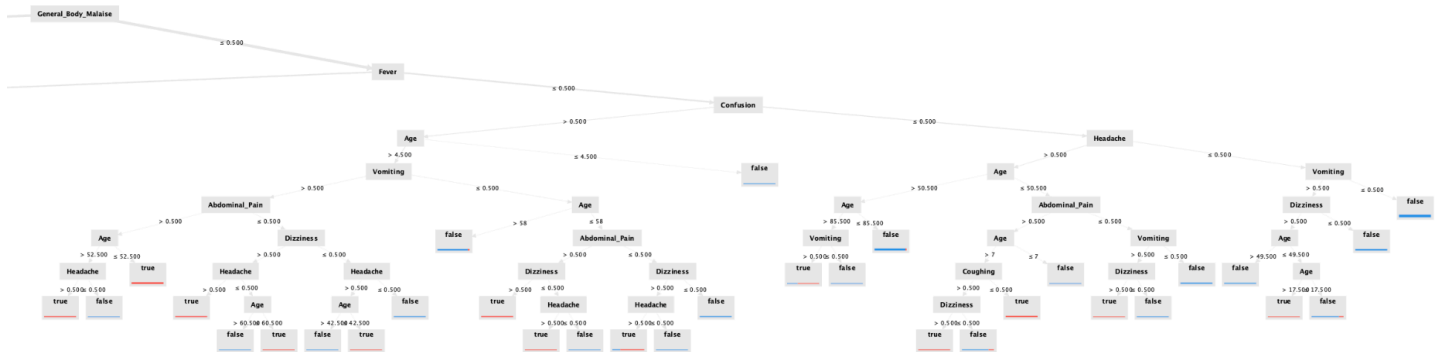
criterion	gain ratio	
maximal depth	10	
☒ apply pruning		
confidence	0.1	
☒ apply prepruning		
minimal gain	0.01	
minimal leaf size	2	
minimal size for split	4	
number of prepruning alternatives	3	

DECISION TREE PARAMETERS: Uses the gain ratio criterion with a maximum depth of 10 applies both pruning (confidence 0.1) and prepruning(minimum gain 0.01, minimum leaf size 2, minimum split size 4, and 3 prepruning alternatives) to control overfitting and improve model accuracy.

Left-Side



Right-Side



DECISION TREE: Visually represents how the model classifies whether a person has malaria (True) or not (False) based on symptom patterns.

ATTRIBUTE WEIGHTS:

attribute	weight
Dizziness	0.125
Headache	0.086
Abdominal_Pain	0.100
Vomiting	0.110
General_Body_Malaise	0.007
Joint_Pain	0.071
Confusion	0.056
Backache	0.109
Coughing	0.117
Fever	0.016
Chest_Pain	0.203

ATTRIBUTE WEIGHTS: Shows the importance of each symptom or attribute in predicting malaria. Higher weights mean greater influence in the model's decision-making.